

**Chapter 02****Articulate Engineering design problem with parameters and needs of perspective stakeholders**

- The economic success of most firms depends on their ability to identify the needs of customers and to quickly create products that meet these needs and can be produced at low cost.
- **A product:** is something sold by an enterprise to its customers.
- **Product development:** is a set of activities beginning with the perception of a market opportunity and ending in the production, sale, and delivery of a product.

**Engineering Design:**

The accreditation board criteria recognized that the process, skills and competencies associated with design are fundamental to practice of engineering. A key feature to a good engineering design is the instilling, gradually but firmly established, of mind set of creative exploration of range of approaches to problems framed as

- complex,
- open-ended,
- Iterative
- multidisciplinary.

The process of making decisions in engineering design requires the use of well-founded skills, competence and knowledge.

**What engineering design is not?**

Engineering design is not being effectively accomplished if the following characteristics are present:

- Immediate or clear solutions
- A single, correct answer
- Solutions relating directly to component specification or sizing.

If the problem requires to specify a size or particular component to accomplish a task, then the design aspect is significantly diminished.

Notably, problems that involve specification and sizing based on standard tables and pre-engineered-type products may be considered more as analysis than design.

**Engineering design component**

Effective engineering design brings together a variety of skills related to design activity and may also involve skills specific to a technical discipline or multiple disciplines as needed. While practitioners bring varied approaches to design as applied to problems within their fields, some overarching characteristics of appropriate design include, but are not limited to:

- development or fostering of creativity
- inclusion of open-ended problems
- development and use of modern design theory and methods
- needs or scope identification
- consideration of constraints such as:
  - health and safety,
  - sustainability,
  - environmental,
  - ethical,
  - security,
  - economics,
  - compliance with regulatory aspects,
  - Universal design issues (including societal, cultural and diversification facets)
  - aesthetics and human factors
- formulation of problem statements and specifications
- consideration of alternative solutions and decision-making
- Feasibility
- Risk analysis
- Production, manufacturing or implementation processes,
- Detailed system description and documentation,
- Testing, prototyping, modelling and validation
- Effective (multi-disciplinary) teamwork and communication skills.

**Characteristics of Successful Product Design and Development:**

From the perspective of the investors in a for-profit enterprise, successful product development results in products that can be *produced and sold profitably*.

However, we could use the following to assess the performance of a product development effort:

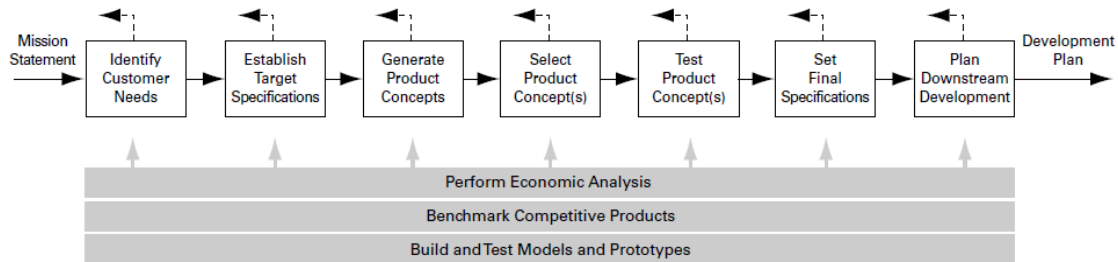
- Product quality: How good is the product resulting from the development effort?
- Product cost: What is the manufacturing cost of the product? This cost includes spending on capital equipment and tooling as well as the incremental cost of producing each unit of the product.
- Development time: How quickly did the team complete the product development effort.
- Development cost: How much did the firm have to spend to develop the product.

**Designs and Develops Products:**

There are three enterprise team central functions for a product development project:

- Marketing: The marketing function mediates the interactions between the firm and its customers.
  - It facilitates the identification of product opportunities,
  - Identification of customer needs.
  - Oversee the launch and promotion of the product.
- Design: The design function plays the lead role in defining the physical form of the product to best meet customer needs.
  - The design function includes engineering design (mechanical, electrical, software, etc.)
  - And industrial design (aesthetics, ergonomics, user interfaces).
- Manufacturing: to coordinate the production system to produce the product. Also, it includes the supply chain; that is
  - Purchasing,
  - Distribution,
  - Installation

## The Generic Engineering design Process:



- The development process shown above is generic,
- Particular processes will differ in accordance with the unique context of the firm and the challenges of any specific project
- The above process used in a market-pull situation: a firm begins product development with a market opportunity and then uses whatever available technologies are required to satisfy the market need (i.e., the market “pulls” the development decisions).

In addition, several variants are common and correspond to the following:

a) Platform Products:

A platform product is built around a pre-existing technological subsystem (a technology platform).

- Examples of such platforms include the Intel chipset in a personal computer, the Apple iPhone operating system, and the blade design in a Gillette razor.
- Products built on technology platforms are much simpler to develop than if the technology were developed from scratch.

b) Process-Intensive Products:

For these products, the production process places strict constraints on the properties of the product

- Examples of process-intensive products include semiconductors, foods, chemicals, and paper.
- In some situations, a new product and new process are developed simultaneously. For example, creating a new shape of breakfast cereal or snack food will require both product and process development activities. In other cases, a specific existing process for making the product is chosen in advance, and the product design is constrained by the capabilities of this process.

c) Customized Products:

Customized products are slight variations of standard configurations and are typically developed in response to a specific order by a customer

- Examples of customized products include switches, motors, batteries, and containers.

d) High-Risk Products

High-risk products are those that involve unusually large uncertainties related to the technology or market so that there is substantial technical or market risk

The Engineering design process addresses many types of risk. These include:

- Technical risk: Will the product function properly,
- Market risk: Will customers like what the team develops,
- Budget and schedule risk: Can the team complete the project on time and within budget.

e) Quick-Build Products

Building and testing prototype models in a rapid process that the design-build-test

- Cycles can be repeated many times.
- Teams can take advantage of rapid iteration to achieve a more flexible and responsive product development process,

f) Complex Systems

Larger-scale products, complex systems comprising many interacting subsystems and components.

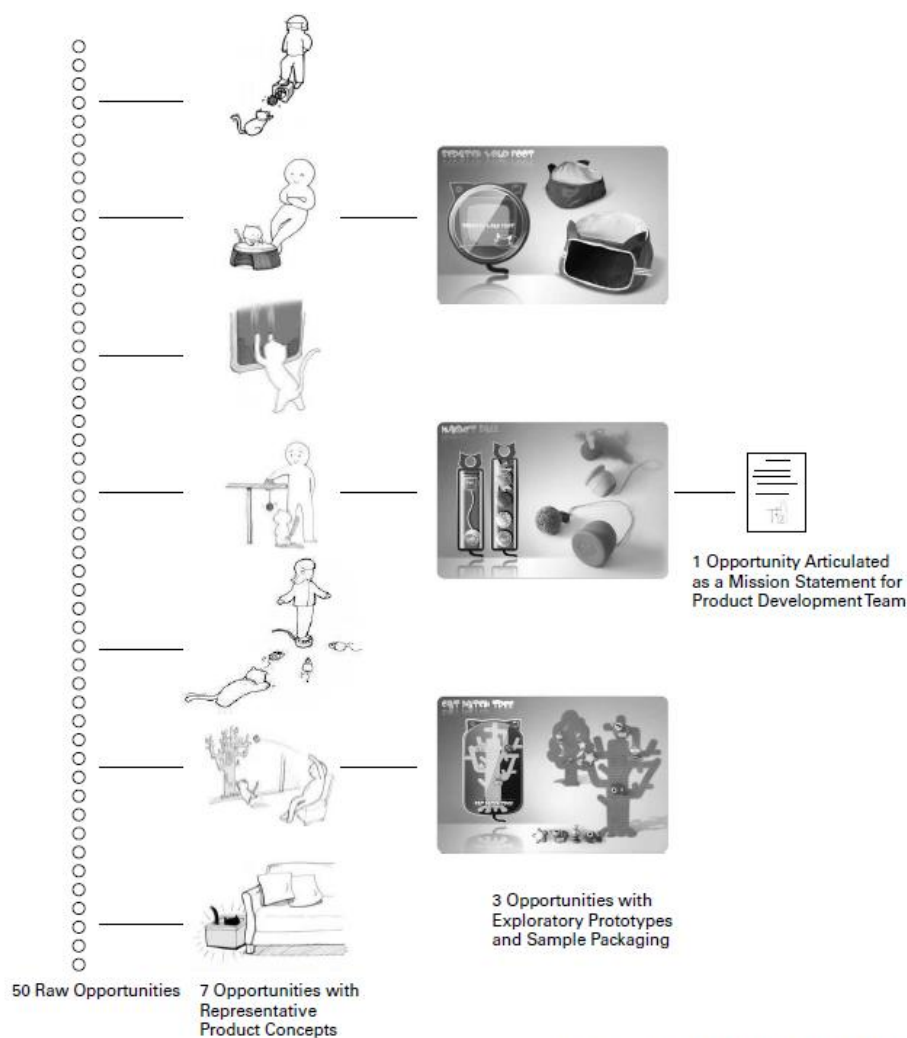
- The concept development phase considers the architecture of the entire system, and multiple architectures may be considered as competing concepts for the overall system.

## Opportunity:

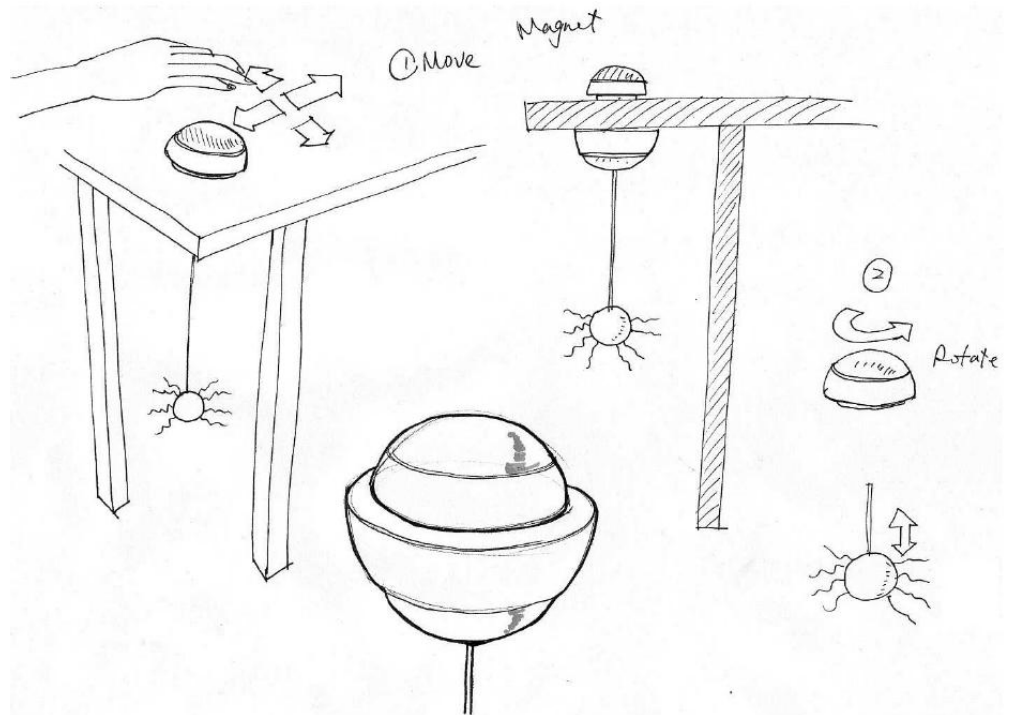
In the setting of product design and development, an opportunity is an idea for a new product. An opportunity is a product description in developing form,

- A newly discovered technology,
- A rough match between a need and a possible solution.

An opportunity for a new product is usually articulated with less than one page of information, often including a descriptive title, a narrative explaining the idea, and sometimes including a sketch of a possible product concept.



*The overall tournament structure of the opportunity identification process for FroliCat. Fifty raw opportunities were eventually filtered and explored, resulting in a “swinging ball” opportunity that was developed into a product launched to the market.*



*The “swinging ball” opportunity eventually pursued by the FroliCat team as first recorded in a sketch. This is an example of an opportunity that includes a potential solution concept.*

### Opportunity Identification Process

Opportunity identification process could be divided into six steps as follows:

1. Establish a charter: Example: the charter for the FroliCat effort was:  
Create a physical product in the cat toy category that we can launch to the market within about a year through our existing retail sales channel
2. Generate and sense many opportunities.
3. Screen opportunities.
4. Develop promising opportunities.
5. Select exceptional opportunities.



*The Sway cat-toy product that resulted from the swinging-ball opportunity.*

### Techniques for Generating Opportunities

The majority of people have a hard time when asked simply to generate some promising opportunities. For them the problem of coming up with something new is simply too abstract, too unstructured, and has too many degrees of freedom.

Following are basic techniques for stimulating the identification of opportunities:

- Follow a Personal Passion:  
List your passions and then consider how emerging technologies, trends, and business models might influence them.
- Compile Bug Lists  
Successful innovators are often constantly dissatisfied with the world around them. They notice unachieved needs of users, including themselves. List (or photograph) every annoyance or frustration you encounter over a period of days or weeks and then pick the most universal and vexing ones and dream up solutions. *Any problem is an opportunity.*
- Study Customers  
Opportunities can be identified by studying customers in a selected market segment. These studies provide a deeper understanding of the true customer needs than you can obtain through surveys.

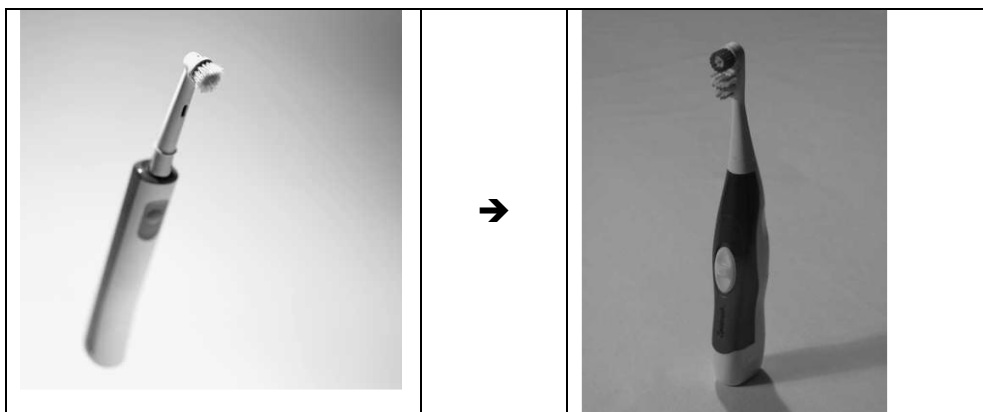


Consider the bicycle industry, a maker of bike components like pedals and brakes, recently commissioned a user-anthropology study to understand why more people in the United States don't ride bikes. The traditional approach to this problem would have been to create a survey or a set of focus groups, asking customers how often they ride and what attributes of a bike they value the most. Most likely, most Americans would say that they ride regularly (which for some might mean once a year) and that they want light bikes with many gears. Those, after all, are the product attributes emphasized in nearly every bike shop.

By spending many hours observing potential cyclists, including time on and off bicycles, researchers found that many consumers want bikes that are technically simple, easy to ride, and easy to get on and off

- Imitate, but Better

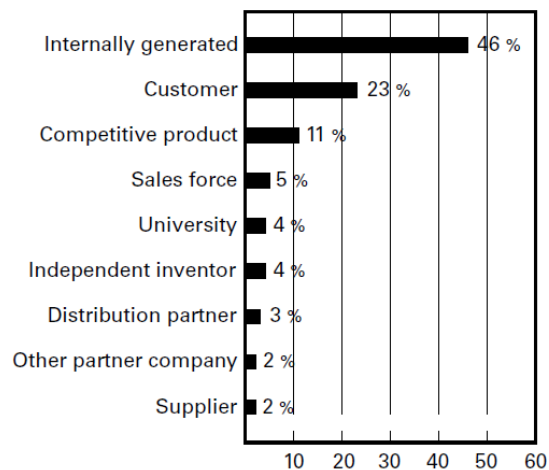
When another firm innovates successfully, you can achieve a new idea by either considering alternative solutions or that could address the same need or alternative needs that could be addressed with the same solution, examples:





### ○ Mine Your Sources

Recall that about half of product opportunities arise from sources inside an organization and about half come from outside sources. As a result, you benefit from cultivating external sources of ideas.



## Product Planning

- The product plan identifies the portfolio of products to be developed by the organization and the timing of their introduction to the market.
- The planning process considers product development opportunities identified by many sources, including suggestions from marketing, research, customers, current product development teams, and benchmarking of competitors

The team needs to know its mission before beginning product development. The answers to these critical questions are included in a mission statement for the team:

- What market segments should be considered in designing the product and developing its features?
- What new technologies (if any) should be incorporated into the new product?
- What are the manufacturing and service goals and constraints?
- What are the financial targets for the project?
- What are the budget and time frame for the project?

## Mission Statements:

The mission statement may include some or all the following information:

- Brief (one-sentence) description of the product: description of the basic function of the product but avoids implying a specific product concept
- Benefit proposition: This element of the mission statement articulates the critical few reasons a customer would buy the product. To some extent this is a hypothesis, which will be validated during the concept development process.
- Key business goals: include goals for time, cost, and quality (e.g., timing of the product introduction, desired financial performance, market share targets).
- Target market(s) for the product: This part of the mission statement identifies the primary market as well as any secondary markets that should be considered in the development effort.
- Assumptions, risks and constraints that guide the development effort: Assumptions must be made carefully; although they restrict the range of possible product concepts, they help to maintain a manageable project scope. Information may be attached to the mission statement to document decisions about assumptions and constraints.

- Stakeholders: One way to ensure that many of the subtle development issues are addressed is to explicitly list all the product's stakeholders, that is, all of the groups of people who are affected by the product's success or failure. Common stakeholders include internal groups like employees and managers, and external parties such as customers, investors, suppliers, government, and communities. The stakeholder list begins with the end user (the ultimate external customer) and the external customer who makes the buying decision about the product. Stakeholders also include customers of the product who reside within the firm, such as the sales force, the service organization, and the production departments. The list of stakeholders serves as a reminder for the team to consider the needs of everyone who will be influenced by the product

Example:

| <b>Mission Statement: Multifunctional Office Document Machine</b> |                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product Description</b>                                        | <ul style="list-style-type: none"> <li>• Networkable, digital machine with copy, print, fax, and scan functions</li> </ul>                                                                                                                                                                                                                                     |
| <b>Benefit Proposition</b>                                        | <ul style="list-style-type: none"> <li>• Multiple document processing functions in one machine</li> <li>• Connected to office computer network</li> </ul>                                                                                                                                                                                                      |
| <b>Key Business Goals</b>                                         | <ul style="list-style-type: none"> <li>• Support Xerox strategy of leadership in digital office equipment</li> <li>• Serve as platform for all future B&amp;W digital products and solutions</li> <li>• Capture 50% of digital product sales in primary market</li> <li>• Environmentally friendly</li> <li>• First product introduction 4th Q 1997</li> </ul> |
| <b>Primary Market</b>                                             | <ul style="list-style-type: none"> <li>• Office departments, mid-volume (40–65 ppm, above 42,000 avg. copies/mo.)</li> </ul>                                                                                                                                                                                                                                   |
| <b>Secondary Markets</b>                                          | <ul style="list-style-type: none"> <li>• Quick-print market</li> <li>• Small "satellite" operations</li> </ul>                                                                                                                                                                                                                                                 |
| <b>Assumptions and Constraints</b>                                | <ul style="list-style-type: none"> <li>• New product platform</li> <li>• Digital imaging technology</li> <li>• Compatible with CentreWare software</li> <li>• Input devices manufactured in Canada</li> <li>• Output devices manufactured in Brazil</li> <li>• Image processing engine manufactured in both the United States and Europe</li> </ul>            |
| <b>Stakeholders</b>                                               | <ul style="list-style-type: none"> <li>• Purchasers and users</li> <li>• Manufacturing operations</li> <li>• Service operations</li> <li>• Distributors and resellers</li> </ul>                                                                                                                                                                               |